

code

effect

`x, y = next(loader)`

fresh shuffled batch

`logits = model(x)`

records graph + activations

`loss = loss_fn(logits, y)`

raw logits in, scalar out

`opt.zero_grad()`

$p.grad \leftarrow \text{None}$

`loss.backward()`

$p.grad \leftarrow \nabla L$, graph freed

`clip_grad_norm(ps, c)`

rescale if $\|g\| > c$, return $\|g\|$

`opt.step()`

update p from $p.grad$ + state

`sched.step()`

set next LR η

`log(loss, gnorm, lr)`

unclipped norm, current LR

next step